

LIDING XU

📍 Berlin, Germany

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EMPLOYMENT

Zuse Institute Berlin, Germany

April 2024 - Present

- Postdoc researcher
- Interactive Optimization and Learning Lab, with Prof. Dr. Sebastian Pokutta

LIX CNRS, École Polytechnique, France

September 2020 - December 2023

- PhD in Computer Science: “Relaxation methods for mixed-integer nonlinear programming”
- Supervisors: Prof. Dr. Leo Liberti, Prof. Dr. Claudia D’Ambrosio, Prof. Dr. Sonia Haddad-Vanier

EDUCATION

Université Paris-Saclay & Institut Polytechnique de Paris, France

September 2019 -

September 2020

- M2 in Applied Mathematics

École Polytechnique, France

September 2018 - August 2019

- M1 in Computer Science

Peking University, China

September 2014 - June 2018

- Bachelor in Machine Intelligence

SUMMARY

Researcher-engineer specializing in mathematical optimization and large-scale solver development. Core developer of **SCIP**, one of the fastest open-source solvers for mixed-integer (nonlinear) programming, with shipped contributions in releases 9.0 and 10.0. Seven years of C/C++ systems programming on performance-critical numerical code, combined with a research record on cutting planes, convexification, and decomposition for MINLP. Recent interests on making LLM agents more effective in optimization tasks and solver tool calls.

PUBLICATIONS

Working papers

- **Projected Parabolic Relaxation of QCQPs.**

Preprints and Submitted Articles

- Liding Xu, Yugeng Zhou, Sebastian Pokutta. **Agentic MIP Research: Accelerated Constraint Handler Generation.** Submitted to *Neurips 2026*.
- Liding Xu, Yechao Liu, and Sebastian Pokutta. 2025. **Convex semidefinite tensor optimization and quantum entanglement.** Submitted to *Mathematical Programming Computation*.
- Liding Xu and Claudia D’Ambrosio. 2025. **A computational framework for continuous edge covering on networks.** Under major revision for *INFORMS Journal on Computing*.

Journal Articles

- Liding Xu and Leo Liberti. 2024. **Relaxations for binary polynomial optimization via signed certificate.** Accepted to *Mathematics of Operations Research*.
- Liding Xu, Claudia D’Ambrosio, Leo Liberti, and Sonia Haddad-Vanier. 2025. **On Cutting Planes for Signomial Programming.** *SIAM Journal on Optimization*.

- Liding Xu and Leo Liberti. 2024. **Submodular Maximization and its Generalization through Intersection Cuts**. *Mathematical Programming*.
- Liding Xu, Claudia D'Ambrosio, Emiliano Traversi, and Sonia Haddad-Vanier. 2023. **Branch and Price for Submodular Bin Packing**. *EURO Journal on Computational Optimization*.
- Mercedes Pelegrín and Liding Xu. 2023. **Continuous Covering on Networks: Improved Mixed Integer Programming Formulations**. *Omega*.
- Liding Xu and Sonia Haddad-Vanier. 2022. **Branch-and-Price for Energy Optimization in Multi-Hop Wireless Sensor Networks**. *Networks*.

Conference Papers

- Liding Xu, Gioni Mexi, and Ksenia Bestuzheva. 2025. **Sparsity-driven Aggregation of Mixed Integer Programs**. In *Proceedings of the 23rd International Symposium on Experimental Algorithms (SEA 2025)*.

Book Chapters

- Antoine Oustry, Liding Xu, Sonia Haddad-Vanier, Juan-Antonio Cordero, and Thomas Clausen. **Optimization in Wireless Networks**. In *Encyclopedia of Optimization*.
- Liding Xu, Claudia D'Ambrosio, Sonia Haddad-Vanier, and Emiliano Traversi. **Urban Air Mobility**. In *Encyclopedia of Optimization*.

Technical Reports

- Suresh Bolusani *et al.* **The SCIP Optimization Suite 9.0**. In *SCIP Optimization Suite Report*.
- Christopher Hojny *et al.* **The SCIP Optimization Suite 10.0**. In *SCIP Optimization Suite Report*.

ENGINEERING EXPERIENCE (OPEN SOURCE)

SCIP — core developer

C

SCIP is a leading open-source solver for MIP, MINLP, used by thousands of academic and industrial users worldwide. Shipped contributions in releases **SCIP 9.0** and **SCIP 10.0** (co-authored release reports).

- Contributed separators and propagators for nonlinear and polynomial constraints, translating peer-reviewed algorithms (signomial cuts, signed-certificate relaxations, subspace inequalities) into production solver code.
- Worked on sparsity-driven aggregation of MIP rows, improving solver performance on benchmark instances (see SEA 2025 paper).

TensorOpt4Entanglement

Julia

Optimization-based tensor methods for quantum entanglement detection; convex semidefinite tensor solver built on top of conic optimization libraries. Companion code to a paper submitted to *Mathematical Programming Computation*.

cflg — continuous set covering on networks

Julia

End-to-end algorithmic toolkit (MIP formulations, preprocessing, cut generation) for continuous edge-covering problems.

cbp — branch-and-price for conic submodular bin packing

C++

Custom column-generation solver with a submodular pricing oracle; supporting code for *EURO Journal on Computational Optimization* paper.

wUMCFC

C++

Solver for wireless unsplittable multi-commodity flow routing with network coding.

SymMIP.jl

Julia

A small symmetry analysis library for mixed-integer programs.

TEACHING

INF569 – Decision theory, with applications to energy systems

École Polytechnique, Palaiseau, France

2022

Master (M1) course – Teaching Assistant (TA)

CSE201 – Object-oriented Programming in C++

École Polytechnique, Palaiseau, France

2022

Bachelor course – Teaching Assistant (TA)

FUNDS AND FELLOWSHIP

Research Campus MODAL funded by the German Federal Ministry of Education and Research (BMBF): 1 year

April 2024 - March 2025

FX-Conseil - Fondation de l'Ecole Polytechnique (Research Foundation of the Ecole Polytechnique): 6 months

October 2023 - March 2024

École doctorale IP Paris (French Ministry of Higher Education and Research): 3 years

September 2020 - September 2024

Labex DigiCosme scholarship for master students: 1 year

September 2018 - August 2019

LANGUAGES

Chinese: Mother Tongue

English: Fluent

French: Beginner

PRESENTATIONS AND ATTENDANCE

ICCOPT2025, Los Angeles, USA

July 2025

Invited talk: Relaxations for Binary Polynomial Optimization via Signed Certificates

EURO2024, Copenhagen, Denmark

July 2024

Invited Talk: Modelling of piece-wise linear concave constraints in continuous covering problems

43e Journée Francilienne de Recherche Opérationnelle: BinPacking workshop

February 2024

Talk: Branch and price for submodular bin packing

Oberwolfach Workshop: Mixed-integer Nonlinear Optimization

August 2023

Attendance

Aussois combinatorial optimization workshop, Aussois, France

January 2023

Talk: Intersection cut meet submodularity

PGMO2022, Paris, France

November 2022

Talk: On a concept of a generic intersection cut callback

SCIP2022, Berlin, Germany

November 2022

Invited Talk: On a concept of a generic intersection cut callback

HUGO2022, Szeged, Hungary

September 2022

Invited Talk: (Three) cutting planes for signomial programming

EURO2022, Espoo, Finland

July 2022

Invited Talk: An algorithmic toolkit for continuous set-covering on networks

EURO2021, Athens, Greece

July 2021

Invited Talk: Optimal location of safety landing sites

ROADEF2021, Mulhouses, France

April 2021

Talk: Optimal location of safety landing sites